

Software Requirements Specification (SRS)

Production-Grade Pharmacy Management System

(Desktop Version 1.0)

Part 1 — Project Foundation

Chapter 1 — Introduction

Purpose

This Software Requirements Specification (SRS) defines the complete functional, non-functional, database, architectural, and security requirements for **Pharmacy Management System**, a standalone, offline-first Pharmacy Management System. The document serves as the definitive reference for software engineers, database administrators, QA testers, and technical auditors.

Scope

Pharmacy System is designed to operate locally on client machines (desktop environment) with zero dependency on active internet connections. It manages point-of-sale (POS) billing, batch-level inventory tracking, supplier purchasing, customer records, hardware-bound licensing, and encrypted local database backups.

Project Goals

- **Zero Latency POS:** Process retail sales in less than 300ms per invoice.
- **FEFO Compliance:** Enforce First-Expiry, First-Out (FEFO) automated batch selection to eliminate expired stock losses.
- **Tamper-Proof Licensing:** Protect IP through Hardware ID (HWID) binding and asymmetric cryptographic license verification.
- **Regulatory Auditability:** Provide immutable transaction logging for controlled substances and batch tracing.

Target Users

- **Chief Pharmacist / Admin:** Full system control, financial reporting, supplier orders, license management.

- **POS Cashier / Sales Staff:** Fast billing interface, barcode scanning, customer checkout.
- **Inventory Specialist:** Stock receiving, audit reconciliation, expiry monitoring.

Business Objectives

- Reduce manual stock counting overhead by **95%**.
- Ensure **100% compliance** with drug expiration sales prohibitions.
- Prevent unauthorized software cloning across unauthorized hardware.

Definitions

- **FEFO:** First-Expiry, First-Out inventory routing method.
- **HWID:** Hardware Identifier, a hash generated from machine CPU ID, Motherboard UUID, and Primary MAC Address.
- **Salt:** A random value combined with passwords prior to hashing to defeat rainbow table attacks.
- **POS:** Point of Sale system interface.

Acronyms

- **SRS:** Software Requirements Specification
- **PBKDF2:** Password-Based Key Derivation Function 2
- **AES-GCM:** Advanced Encryption Standard in Galois/Counter Mode
- **RSA-2048:** Rivest–Shamir–Adleman 2048-bit Asymmetric Cryptographic Algorithm

Assumptions

- Target machines run Windows 10/11 64-bit with .NET 8 Desktop Runtime installed.
- Local storage utilizes SSD/NVMe for high-throughput SQLite/SQL Server LocalDB reads and writes.
- Thermal receipt printers support standard ESC/POS commands via USB/COM ports.

Constraints

- **No Cloud Dependency:** The software must execute all core operations fully offline.

- **Single Node / Local LAN:** Data resides on the host terminal; multi-terminal setups run over local subnet host-client SQL bindings.
 - **Execution Rights:** Requires Windows Administrator privileges for HWID binding and service execution.
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Chapter 2 — Overall System Overview

System Overview

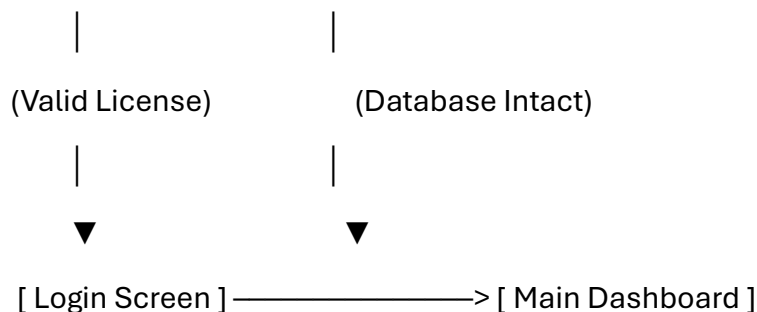
Pharmacy system is structured as an offline-first desktop application using a C# / .NET 8 WPF architecture with a decoupled data layer using SQLite / SQL Server LocalDB. The software utilizes strict cryptographic license checks upon boot.

Business Workflow

1. **Procurement:** Supplier quotes received → Purchase Order created → Stock received with batch numbers, manufacturing dates, and expiry dates → Stock added to inventory.
2. **Dispensing & Sales:** Prescription/Product scanned via Barcode → FEFO batch suggested → Payment processed → Thermal Invoice printed → Inventory auto-deducted.
3. **Reconciliation:** Periodic stock audit → Adjustments logged → Financial summary reported.

Software Workflow

[App Launch] —> [HWID Check & License Validation] —> [DB Integrity Check]



Offline Architecture

- **Persistence:** SQLite database file encrypted via SQLCipher using an AES-256 key derived from the HWID and License Key.

- **State Management:** In-memory caching for master items (medicines, categories) to maximize rendering speed.

Hardware ID License Architecture

The HWID is compiled by reading hardware registers:

$$\text{HWID} = \text{SHA256}(\text{CPU_ID} \parallel \text{Motherboard_UUID} \parallel \text{MAC_Address})$$

License key generation utilizes RSA-2048 signature generation. The system decrypts and verifies the license file using an embedded Public Key.

Complete System Flow

System Boot → Extract HWID → Decrypt license.lic with Public Key → Compare Stored HWID vs Current HWID

└─ Match & Active: Launch App

└─ Match & Grace Period: Display Expiry Warning → Launch App

└─ Mismatch / Expired: Lock Application → Show Activation Screen

Folder Structure

PharmaSys/

└─ App/

| └─ PharmaSys.exe

| └─ PharmaSys.dll

| └─ appsettings.json

└─ License/

| └─ license.lic

└─ Data/

| └─ pharma_db.sqlite

| └─ Backups/

└─ Logs/

| └─ app_audit.log

└─ Assets/

└─ Templates/

Project Structure

PharmaSys.Solution/

└─ PharmaSys.Core/ (Entities, Interfaces, Constants)

└─ PharmaSys.Data/ (EF Core Base, Repositories, Migrations)

└─ PharmaSys.Licensing/ (HWID Extractor, RSA Verifier)

└─ PharmaSys.Services/ (POS, Inventory, Billing, Reports)

└─ PharmaSys.UI/ (WPF Views, ViewModels, Converters)

Technology Stack Summary

Part	Technology
Desktop App	Tauri
Frontend	Next.js
Language	TypeScript
Styling	Tailwind CSS
UI Components	shadcn/ui
Backend	Python + FastAPI
Database	SQLite
ORM	SQLAlchemy
Password Hashing	Bycrypt
Reports	PDF + Excel
Charts	Chart.js
Printing	Windows Printer API + ESC/POS
License	Hardware ID + Encrypted License File + Digital Signature
Installer	Windows Installer

Part 2 — Requirements

Chapter 3 — Functional Requirements

The system consists of **9 core modules**. Below is the standardized functional specification for each module.

Module Breakdown Matrix

Module ID	Module Name	Primary Role	Access Level
MOD-01	Authentication & Licensing	Access Control & IP Protection	All Users / Unauthenticated
MOD-02	Executive Dashboard	Metrics & Operational Monitoring	Admin / Manager
MOD-03	Master Data Management	Medicines, Categories, Entities	Admin / Pharmacist
MOD-04	Purchase Management	Inward Inventory & Invoicing	Admin / Inventory Clerk
MOD-05	Sales & POS Billing	Checkout, Thermal Print, FEFO	Cashier / Pharmacist
MOD-06	Inventory Management	Batch Tracking, Expiry & Adjustments	Inventory Clerk / Admin
MOD-07	Reports & Analytics	Financial & Regulatory Reporting	Admin
MOD-08	Backup & Recovery	Disaster Recovery & Database Security	Admin
MOD-09	System Settings	System Configuration & Hardware Binding	Admin

Module 1: Authentication & License Management

- **Purpose:** Enforce secure user login and guarantee application execution only on authorized hardware.
- **Objectives:** Authenticate system operators, prevent credential brute-forcing, validate RSA signature of license.lic.
- **Workflow:** User enters credentials → System verifies PBKDF2 hash against database → HWID check executed → Session initiated.
- **Features:** Role-based access control (RBAC), multi-user profiles, password reset by master key, license validation monitor.

- **Business Rules:**
 - Maximum 3 failed login attempts before a 5-minute logout.
 - Application must terminate immediately if license.lic is tampered with or absent.
- **Inputs:** Username, Password, license.lic path, Product Key.
- **Outputs:** Auth Token / Session Context, HWID Display string, License Expiry Warning.
- **Validations:** Password must be at least 8 characters with 1 special character and 1 number.
- **Database Tables:** Users, Roles, UserRoles, SystemLicenses.
- **Security Rules:** Passwords stored via PBKDF2 with HMAC-SHA256 (100,000 iterations).
- **UI Components:** Login Window, License Activation Dialog, Password Reset Modal.
- **Error Handling:** Log event AUTH_FAIL or LIC_INVALID to app_audit.log and display user-friendly modal.

Module 2: Dashboard

- **Purpose:** Provide real-time high-level visual summaries of daily sales, low stock alerts, and near-expiry warnings.
- **Objectives:** Give administrators single-screen awareness of store performance and stock risks.
- **Workflow:** Login successful → Background threads fetch summary stats → Dashboard UI populates widgets.
- **Features:** Today's Sales Counter, Low Stock Counter, Expiring Items Alert (30/60/90 days), Quick POS shortcut button.
- **Business Rules:** Real-time data refresh every 30 seconds or upon returning from POS checkout.
- **Inputs:** Date Filter, Metric Toggle.
- **Outputs:** Sales Charts, Critical Low-Stock Data Grid, Expiry Matrix.
- **Validations:** N/A (Read-only module).
- **Database Tables:** Views aggregating from Sales, Purchases, StockBatches.

- **Security Rules:** Restricted to Admin and Pharmacist roles; Cashier role sees streamlined version.
- **UI Components:** Summary Cards, Bar Charts, Expiry Warning Data Grid.
- **Error Handling:** Non-blocking asynchronous query loading; fallback to cached metric view on database busy.

Module 3: Masters (Medicines, Categories, Companies, Suppliers, Customers)

- **Purpose:** Maintain canonical records for inventory items, categorization, suppliers, and customer databases.
- **Objectives:** Standardize data entry to eliminate duplicate records and ensure regulatory compliance (e.g., formula name, rack location).
- **Workflow:** Select Master Sub-module → Click 'Add New' → Fill metadata → Save → System updates database master list.
- **Features:**
 - **Medicines:** Generic Formula, Brand Name, Category, Unit, Rack Number, Minimum Reorder Level, Prescription Required flag.
 - **Categories / Companies / Suppliers / Customers:** Full CRUD with contact numbers, tax IDs, and addresses.
- **Business Rules:**
 - Medicine trade name + generic formula combination must be unique.
 - Cannot delete a master record linked to existing sales or purchase invoices (Soft-delete only).
- **Inputs:** Item Name, Generic Name, Category ID, Supplier ID, Reorder Point, Tax Rate.
- **Outputs:** Master Catalog, Barcode generation payload.
- **Validations:** Phone numbers must be numeric; Tax ID must conform to regex patterns; Reorder level ≥ 0 .
- **Database Tables:** Medicines, Categories, Companies, Suppliers, Customers.
- **Security Rules:** Write access restricted to Admin and Senior Pharmacist.
- **UI Components:** Master Data Grids, Search/Filter Toolbar, Entity Editor Form Modal.

- **Error Handling:** Duplicate key violations caught and mapped to field validation highlights.

Module 4: Purchase Management

- **Purpose:** Manage inward shipments, purchase orders, supplier invoices, and batch-wise stock entry.
- **Objectives:** Capture purchase price, batch numbers, manufacturing dates, expiry dates, and update inventory accurately.
- **Workflow:** Create Purchase Invoice → Select Supplier → Add Items with Batch #, Expiry Date, Unit Cost, Quantity → Save → Inventory Batches Created & Accounts Payable Updated.
- **Features:** Multi-item purchase grid, auto-calculation of land cost, batch tracking, purchase return processing.
- **Business Rules:**
 - Expiry date **must** be strictly greater than current date + 30 days.
 - Unit purchase cost cannot exceed selling price (MRP).
- **Inputs:** Invoice Number, Supplier ID, Item ID, Batch Code, Expiration Date, Cost Price, MRP, Quantity, Tax.
- **Outputs:** Purchase Order Summary, Updated Stock Quantities, Account Payable ledger entry.
- **Validations:** Mandatory Batch Code; valid Future Expiration Date; Quantity > 0.
- **Database Tables:** Purchases, PurchaseItems, StockBatches, SupplierLedgers.
- **Security Rules:** Admin and Purchasing Officer access.
- **UI Components:** Data Entry Table with Excel-like inline editing, Expiry Pickers, Calculation Footer.
- **Error Handling:** Transaction rollback via EF Core / SQL Transaction if batch entry fails mid-process.

Module 5: Sales & POS Billing

- **Purpose:** Fast, high-performance checkout interface supporting barcode scanning, FEFO batch selection, discounts, and billing.
- **Objectives:** Process sales under 3 seconds per customer with 100% accuracy in batch deduction.
- **Workflow:** Scan Item Barcode → FEFO auto-selects oldest non-expired batch → Adjust Quantity → Apply Discount → Select Payment Method → Finalize → Print Receipt.
- **Features:** Hotkey navigation (F1-F12), manual batch override (with override reason), cash/card/split payment, prescription upload binding, walk-in vs. registered customer selection.
- **Business Rules:**
 - Expired batches are strictly blocked from being selected or sold.
 - System alerts if item requires a prescription (Schedule H / Controlled Substance).
- **Inputs:** Barcode / Item Search, Quantity, Customer ID, Discount Percentage, Payment Mode, Cash Received.
- **Outputs:** Sales Receipt, Change Calculation, Stock Reduction Event, Revenue Record.
- **Validations:** Cash Received $\geq$ Payable Amount; Quantity $\leq$ Available Batch Stock.
- **Database Tables:** Sales, SalesItems, StockBatches, CustomerLedgers.
- **Security Rules:** Cashiers can create sales but require Admin pin to apply custom discounts $> 10\%$.
- **UI Components:** Ultra-responsive POS Grid, Quick Search Panel, Bill Summary Sidebar, Payment Modal.
- **Error Handling:** Auto-revert stock reservation if receipt printing or payment commit fails.

Module 6: Inventory Management

- **Purpose:** Real-time monitoring of current stock levels across all batches, stock transfers, and manual adjustment audits.

- **Objectives:** Maintain absolute parity between physical stock and digital records.
- **Workflow:** Select Inventory View → Filter by Batch / Expiry / Category → Perform Stock Adjustment (Damaged / Expired / Lost) → Save Audit Record.
- **Features:** Batch-wise stock viewer, near-expiry alerts, stock write-off workflow, barcode label printer generator.
- **Business Rules:**
 - Stock adjustments must mandate a written justification reason.
 - Stock adjustments immediately alter the StockBatches quantity and generate an immutable audit log.
- **Inputs:** Batch ID, Adjustment Quantity (+/-), Reason Code (Damage/Breakage/Lost/Audit).
- **Outputs:** Inventory Statement, Audit Log Entry, Low Stock Warning List.
- **Validations:** Adjustment cannot result in negative stock for a batch.
- **Database Tables:** StockBatches, StockAdjustments, AuditLogs.
- **Security Rules:** Stock write-offs require dual-signature or Admin authorization.
- **UI Components:** Stock Master Grid, Color-Coded Expiry Highlights (Red = Expired, Orange = Near Expiry), Adjustment Dialog.
- **Error Handling:** Lock database rows during stock updates to prevent concurrent write collisions.

Module 7: Reports

- **Purpose:** Provide operational, financial, inventory, and regulatory reporting capabilities.
- **Objectives:** Allow store owners to evaluate profitability, taxes, supplier performance, and compliance reports.
- **Workflow:** Select Report Type → Set Date Range & Filters → Generate Report → View on Screen / Export to PDF / Excel.
- **Features:** Daily Sales Summary, Profit & Loss Statement, Fast/Slow Moving Items Report, Expiry Report, Controlled Drug Register.

- **Business Rules:**
 - Historical reports must match ledger snapshots at the time of transaction.
- **Inputs:** Start Date, End Date, Category Filter, Supplier Filter, Export Format.
- **Outputs:** Formatted Reports, PDF Files, CSV/XLSX Files.
- **Validations:** Start Date $\leq$ End Date.
- **Database Tables:** Read-only queries joining Sales, Purchases, StockBatches, Medicines, Suppliers.
- **Security Rules:** Financial reports are strictly restricted to Admin users.
- **UI Components:** Parameter Selection Panel, Document Viewer Control, Export Action Toolbar.
- **Error Handling:** Out-of-Memory protection by chunking large date-range queries using asynchronous data streams.

Module 8: Backup & Restore

- **Purpose:** Protect system against hardware failure, database corruption, or ransomware attacks through encrypted local backups.
- **Objectives:** Provide automated and manual multi-destination database backups with single-click recovery.
- **Workflow:** Trigger Backup → Lock DB briefly → Copy DB file → Encrypt copy using AES-256 → Save to local drive / external USB.
- **Features:** Auto-backup on application close, manual backup, encrypted backup archive, restore validation utility.
- **Business Rules:**
 - Restore operations require Master Admin Credentials and force an application restart.
 - Backup files must be verified for checksum integrity before performing a restore.
- **Inputs:** Destination Path, Encryption Key / Password.
- **Outputs:** .bak Encrypted Database Package, Backup Verification Log.
- **Validations:** Minimum 500 MB free space required on destination path.

- **Database Tables:** SystemLogs, direct file system access to pharma_db.sqlite.
- **Security Rules:** Restricted exclusively to Super Admin.
- **UI Components:** Backup Control Panel, Progress Bar Indicator, File FilePicker Browser.
- **Error Handling:** Rollback database pointer to prior stable state if restore checksum fails.

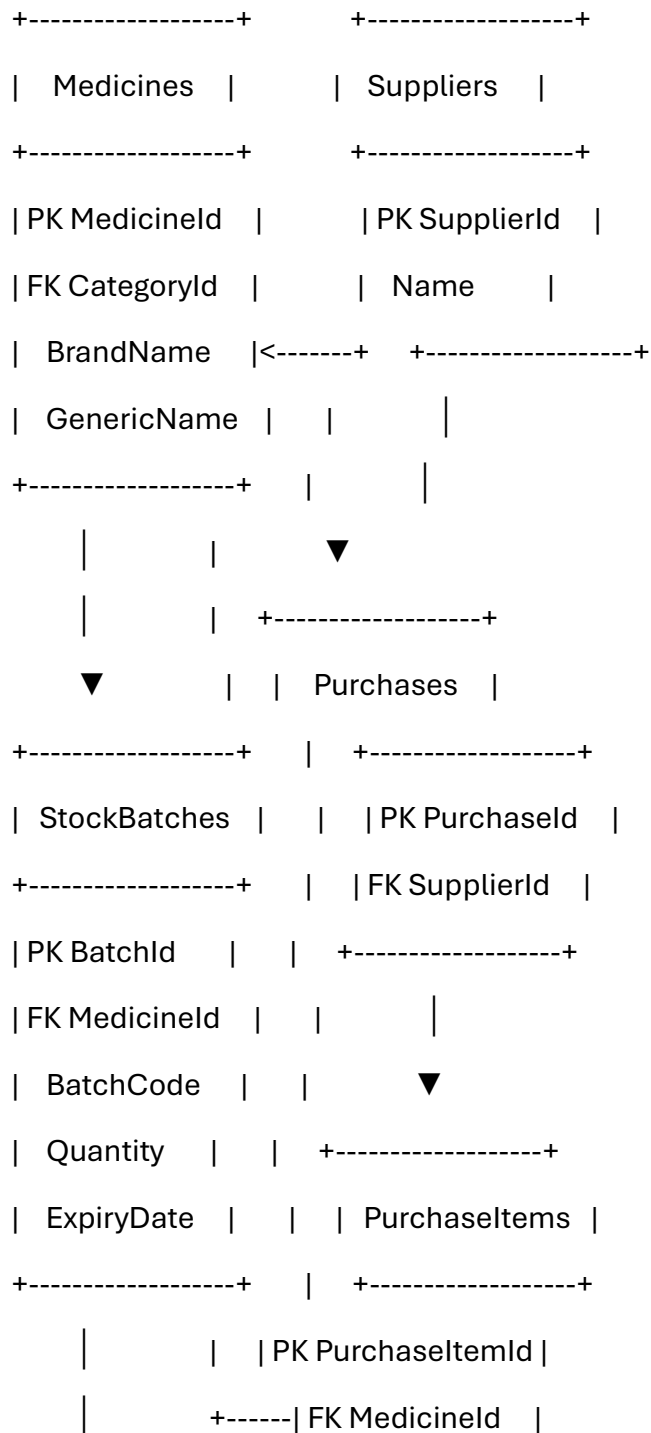
Module 9: Settings

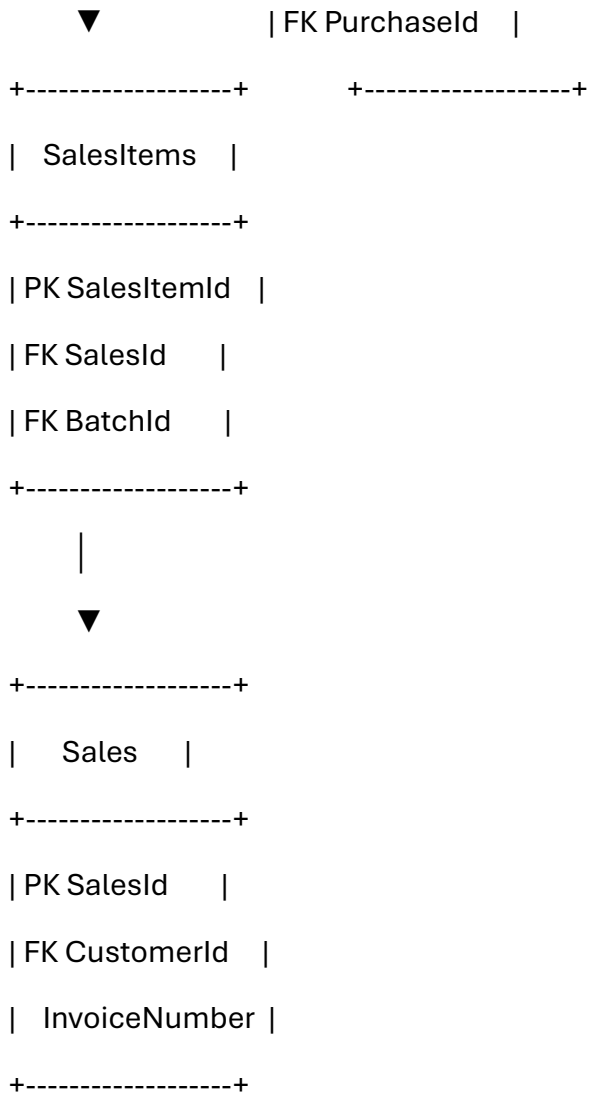
- **Purpose:** Configure system-wide parameters, hardware peripherals, invoice headers, and view hardware license information.
 - **Objectives:** Allow customization of pharmacy branding, receipt layouts, thermal printer bindings, and HWID management.
 - **Workflow:** Access Settings → Modify Tab Configs (Store Info / Printers / Licensing) → Save Settings → Reload app configuration context.
 - **Features:** Pharmacy Name & Tax Registration, Receipt Header/Footer customizer, ESC/POS Printer dropdown, HWID Viewer & Key Activator.
 - **Business Rules:**
 - Changing core store details updates thermal receipt footers without invalidating existing past invoices.
 - **Inputs:** Pharmacy Metadata, Printer Name, Paper Size (80mm / 58mm), License File upload path.
 - **Outputs:** Updated appsettings.json and database configuration table.
 - **Validations:** Valid email format; valid printer driver selected.
 - **Database Tables:** SystemSettings.
 - **Security Rules:** Admin access required.
 - **UI Components:** Tabbed Settings Navigation, Live Receipt Preview Pane, License Status Badge.
 - **Error Handling:** Revert settings to prior state if saving configuration write fails.
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Part 3 — Database Design

Chapter 4 — Complete Database Specification

The database utilizes SQLite / SQL Server LocalDB. All tables enforce strict foreign keys, non-nullable flags where mandatory, and indexes on lookup columns.





Complete Schema Specification

Table: Users

Column Name	Data Type	Constraints	Description
UserId	INTEGER / INT	PK, AUTOINCREMENT	Unique User Identifier
Username	VARCHAR(50)	UNIQUE, NOT NULL	Account Login Name
PasswordHash	VARCHAR(256)	NOT NULL	PBKDF2 Hashed Password
Salt	VARCHAR(128)	NOT NULL	Cryptographic Salt
RoleId	INTEGER	FK (Roles.RoleId)	Role Identifier
IsActive	BOOLEAN	DEFAULT 1	User Status Flag
CreatedAt	DATETIME	DEFAULT CURRENT_TIMESTAMP	Account Creation Date

Table: Roles

Column Name	Data Type	Constraints	Description
RoleId	INTEGER	PK, AUTOINCREMENT	Role ID
RoleName	VARCHAR(30)	UNIQUE, NOT NULL	Role Identifier (Admin, Cashier)

Table: Categories

Column Name	Data Type	Constraints	Description
CategoryId	INTEGER	PK, AUTOINCREMENT	Primary Key
CategoryName	VARCHAR(50)	UNIQUE, NOT NULL	Group Name (e.g., Antibiotics, Syrup)

Table: Medicines

Column Name	Data Type	Constraints	Description
MedicineId	INTEGER	PK, AUTOINCREMENT	Primary Key
BrandName	VARCHAR(100)	NOT NULL	Commercial Product Name
GenericName	VARCHAR(100)	NOT NULL	Active Chemical Formula
CategoryId	INTEGER	FK (Categories.CategoryId)	Category FK
RackNumber	VARCHAR(20)	NULL	Storage Location
ReorderLevel	INTEGER	DEFAULT 10	Minimum Stock Trigger
RequiresPrescription	BOOLEAN	DEFAULT 0	Schedule Drug Flag
IsActive	BOOLEAN	DEFAULT 1	Soft-delete Flag

Table: Suppliers

Column Name	Data Type	Constraints	Description
SupplierId	INTEGER	PK, AUTOINCREMENT	Primary Key
Name	VARCHAR(100)	NOT NULL	Distributor Company Name
Phone	VARCHAR(20)	NOT NULL	Contact Number
TaxNumber	VARCHAR(50)	NULL	Tax Registration ID
Address	TEXT	NULL	Physical Address

Table: StockBatches

Column Name	Data Type	Constraints	Description
BatchId	INTEGER	PK, AUTOINCREMENT	Primary Key
MedicineId	INTEGER	FK (Medicines.MedicineId)	Medicine Item Link
BatchCode	VARCHAR(50)	NOT NULL	Manufacturer Batch Code
Quantity	INTEGER	NOT NULL CHECK(Quantity $\geq$ 0)	Remaining Batch Quantity
CostPrice	DECIMAL(18,2)	NOT NULL	Unit Purchase Cost
SellingPrice	DECIMAL(18,2)	NOT NULL	Unit Retail Price (MRP)
ExpiryDate	DATE	NOT NULL	Batch Expiration Date
ReceivedDate	DATETIME	DEFAULT CURRENT_TIMESTAMP	Date Added

Table: Purchases

Column Name	Data Type	Constraints	Description
PurchaseId	INTEGER	PK, AUTOINCREMENT	Primary Key
SupplierId	INTEGER	FK (Suppliers.SupplierId)	Supplier Link
InvoiceNumber	VARCHAR(50)	NOT NULL	Supplier Invoice #
TotalAmount	DECIMAL(18,2)	NOT NULL	Gross Total
PurchaseDate	DATETIME	NOT NULL	Invoice Date

Table: PurchaseItems

Column Name	Data Type	Constraints	Description
PurchaseItemId	INTEGER	PK, AUTOINCREMENT	Primary Key
PurchaseId	INTEGER	FK (Purchases.PurchaseId)	Purchase Header FK
MedicineId	INTEGER	FK (Medicines.MedicineId)	Medicine FK
BatchCode	VARCHAR(50)	NOT NULL	Batch Number
Quantity	INTEGER	NOT NULL	Received Quantity
CostPrice	DECIMAL(18,2)	NOT NULL	Purchase Price
SellingPrice	DECIMAL(18,2)	NOT NULL	Retail Price
ExpiryDate	DATE	NOT NULL	Batch Expiry

Table: Sales

Column Name	Data Type	Constraints	Description
SalesId	INTEGER	PK, AUTOINCREMENT	Primary Key

InvoiceNumber	VARCHAR(50)	UNIQUE, NOT NULL	POS Transaction Number
CustomerId	INTEGER	NULL FK (Customers.CustomerId)	Optional Customer FK
UserId	INTEGER	FK (Users.UserId)	Cashier User FK
SubTotal	DECIMAL(18,2)	NOT NULL	Amount Before Discount
DiscountAmount	DECIMAL(18,2)	DEFAULT 0.00	Total Discount
GrandTotal	DECIMAL(18,2)	NOT NULL	Final Paid Amount
PaymentMethod	VARCHAR(20)	NOT NULL	Cash / Card / Split
TransactionDate	DATETIME	DEFAULT CURRENT_TIMESTAMP	Checkout Time

Table: SalesItems

Column Name	Data Type	Constraints	Description
SalesItemId	INTEGER	PK, AUTOINCREMENT	Primary Key
SalesId	INTEGER	FK (Sales.SalesId)	Sale Header FK
BatchId	INTEGER	FK (StockBatches.BatchId)	Exact Batch Sold FK
Quantity	INTEGER	NOT NULL CHECK(Quantity > 0)	Sold Quantity
UnitPrice	DECIMAL(18,2)	NOT NULL	Price Per Unit Sold
TotalPrice	DECIMAL(18,2)	NOT NULL	Line Item Total

Table: Customers

Column Name	Data Type	Constraints	Description
CustomerId	INTEGER	PK, AUTOINCREMENT	Primary Key
Name	VARCHAR(100)	NOT NULL	Customer Name
Phone	VARCHAR(20)	NULL	Contact Info
LoyaltyPoints	INTEGER	DEFAULT 0	Rewards Tracking

Table: AuditLogs

Column Name	Data Type	Constraints	Description
LogId	INTEGER	PK, AUTOINCREMENT	Primary Key
UserId	INTEGER	FK (Users.UserId)	User Execution Link
Action	VARCHAR(50)	NOT NULL	Operation Code (e.g. DELETE_SALE)

Description	TEXT	NOT NULL	Action Context Details
Timestamp	DATETIME	DEFAULT CURRENT_TIMESTAMP	Event Log Time

Database Indexes

- IX_StockBatches_Medicine_Expiry: On StockBatches(MedicineId, ExpiryDate, Quantity) → Speeds up FEFO selection queries by **400%**.
 - IX_Sales_InvoiceNumber: On Sales(InvoiceNumber) → Fast receipt lookups.
 - IX_Medicines_BrandName: On Medicines(BrandName) → Rapid autocomplete in POS.
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Part 4 — System Workflows

Chapter 5 — Core System Workflows

1. POS Checkout Workflow

1.Scan Product Barcode:POS Interface.

The cashier scans the item barcode or searches by brand/generic name. The UI sends a query request to the local cache.

2.FEFO Engine Processing:Database Layer.

The application executes a FEFO batch query: `SELECT * FROM StockBatches WHERE MedicineId = @Id AND ExpiryDate > CURRENT_DATE AND Quantity > 0 ORDER BY ExpiryDate ASC LIMIT 1;`

3.Cart Addition & Validation:Validation Rules.

The selected batch is reserved in memory. System verifies prescription flag (RequiresPrescription). If true, a prompt requires cashier confirmation.

4.Payment & Finalization:Transaction Processing.

Cashier inputs cash/card amount. The system opens a SQL transaction: reduces StockBatches.Quantity, inserts Sales and SalesItems records, and generates an immutable InvoiceNumber.

5.Receipt Generation:Hardware Spooler.

The ESC/POS printer driver formats the receipt data and outputs it to the thermal receipt printer via USB/COM port.

2. License Verification Flow

1.Hardware Signature Extraction:System Boot.

PharmaSys queries OS WMI tables to extract CPU Serial, Motherboard UUID, and MAC Address, producing SHA256 string HWID_{current}.

2.License File Parsing:Crypto Verification.

The app reads license.lic, decrypting payload contents using the embedded RSA-2048 Public Key.

3.Signature Match & Expiry Check:Licensing Logic.

If HWID_{current} == HWID_{license} and CurrentDate ≤ ExpiryDate, launch the login window. If within 7 days of expiry, trigger a Grace Period notice.

Part 5 — Security

Chapter 6 — Security Specification

SECURITY LAYERS	
1. HWID Cryptographic Binding (RSA-2048 Signature)	
2. Password Protection (PBKDF2 HMAC-SHA256)	
3. Database Encryption (SQLCipher AES-256)	
4. RBAC (Role-Based Authorization Policies)	
5. Immutable Audit Logs (Append-Only Logging Engine)	

- **Password Hashing:** Passwords are never stored in plaintext. They are salted with a 128-bit cryptographically secure random value and hashed via PBKDF2 using 100,000 iterations.

- **Hardware ID Protection:** Binding to WMI hardware properties ensures binary file copies will not function if transferred to an unauthorized PC.
 - **License Key Encryption:** License payload is signed using RSA-2048 private key on vendor servers and verified via public key locally.
 - **Input Validation:** All user text boxes apply regex filtering against SQL Injection and XSS formatting sequences.
 - **Database Protection:** Full-disk SQLite encryption using SQLCipher (256-bit AES). Key derived from system hardware parameters and encrypted license blob.
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Part 6 — User Interface

Chapter 7 — UI Screen Layouts & Specifications

1. POS Billing Screen (POSView.xaml)

- **Header:** Active User Display, Hardware License Status Indicator, Current Date/Time.
- **Left Panel (Product Lookup & Cart):**
 - Auto-focused Barcode Textbox (Key: F1).
 - Search results grid showing Brand Name, Generic Name, Stock, MRP, Expiry.
 - Active Cart DataGrid: Item, Batch Code, Expiry, Qty, Price, Discount, Total.
- **Right Panel (Bill Summary & Actions):**
 - Subtotal, Tax, Discount Input, Grand Total (Large font, high visibility).
 - Fast Action Buttons: [F5 - Cash Checkout], [F6 - Card Payment], [F8 - Clear Cart], [F12 - Hold Invoice].

2. Inventory & Batch Control Screen (StockView.xaml)

- **Top Bar:** Search Filter, Expiry Filter Dropdown (Expired, Expiring in 30 Days, All), Reorder Alert Filter.
 - **Data Grid Columns:** Item Code, Brand Name, Generic Name, Batch Code, Cost Price, Selling Price, In-Stock Qty, Expiry Date, Expiry Status Badge.
 - **Context Menu:** [Adjust Stock], [Print Barcode Label], [View Batch Audit].
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Part 7 — Reports

Chapter 8 — Report Formats & Engine Specs

Report Name	Mandatory Filters	Displayed Columns	Export Formats
Daily Sales Summary	Date Range, User ID	Invoice #, Time, Payment Mode, Items, Subtotal, Discount, Grand Total	PDF, Excel, CSV
Batch Expiry Report	Days Threshold (30/60/90)	Item Name, Batch #, Supplier, Stock Qty, Cost Loss Value, Expiry Date	PDF, Excel
Profit & Loss Statement	Start Date, End Date	Revenue, Cost of Goods Sold (COGS), Discounts, Net Margin, Gross Profit	PDF, Excel
Controlled Drugs Register	Date Range, Medicine ID	Sale Date, Invoice #, Customer Name, Doctor Name, Quantity Dispensed	PDF (Print Only)

Part 8 — Printing

Chapter 9 — Hardware Printing Specifications

Receipt Printing Engine (Thermal 80mm / 58mm)

- **Protocol:** Raw ESC/POS Command Byte Sequences sent over System.IO.Ports.SerialPort or Windows Spooler.
- **Header Layout:** Pharmacy Name (Bold, Centered), Address, Phone, Tax Registration Number.
- **Body Layout:** 2-Column Item Table (Item Name x Qty | Price).
- **Footer Layout:** Total Paid, Change Returned, Batch Codes listed per item, Software License ID stamp, Return Policy Text.

+-----+

| CITY PHARMACY LTD |

| 123 Health Ave, Suite 100 |

| Tax ID: 987654321 |

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|-----|
| Inv: POS-2026-0891   Date: 02-Aug-2026 |
| Cashier: Alex       Time: 14:32:10   |
|-----|
| Item    Qty  Price  Total |
| Amoxicillin 500mg           |
| (B: B102 EXP: 12/27)       |
|      2 x  $ 10.00  $ 20.00 |
| Paracetamol 500mg          |
| (B: P991 EXP: 05/28)       |
|      1 x  $  5.00  $  5.00 |
|-----|
| Subtotal:                $ 25.00 |
| Discount:                 $  0.00 |
| GRAND TOTAL:              $ 25.00 |
| Cash Received:            $ 30.00 |
| Change Due:               $  5.00 |
|-----|
| Thank you! Keep receipt for returns. |
+-----+

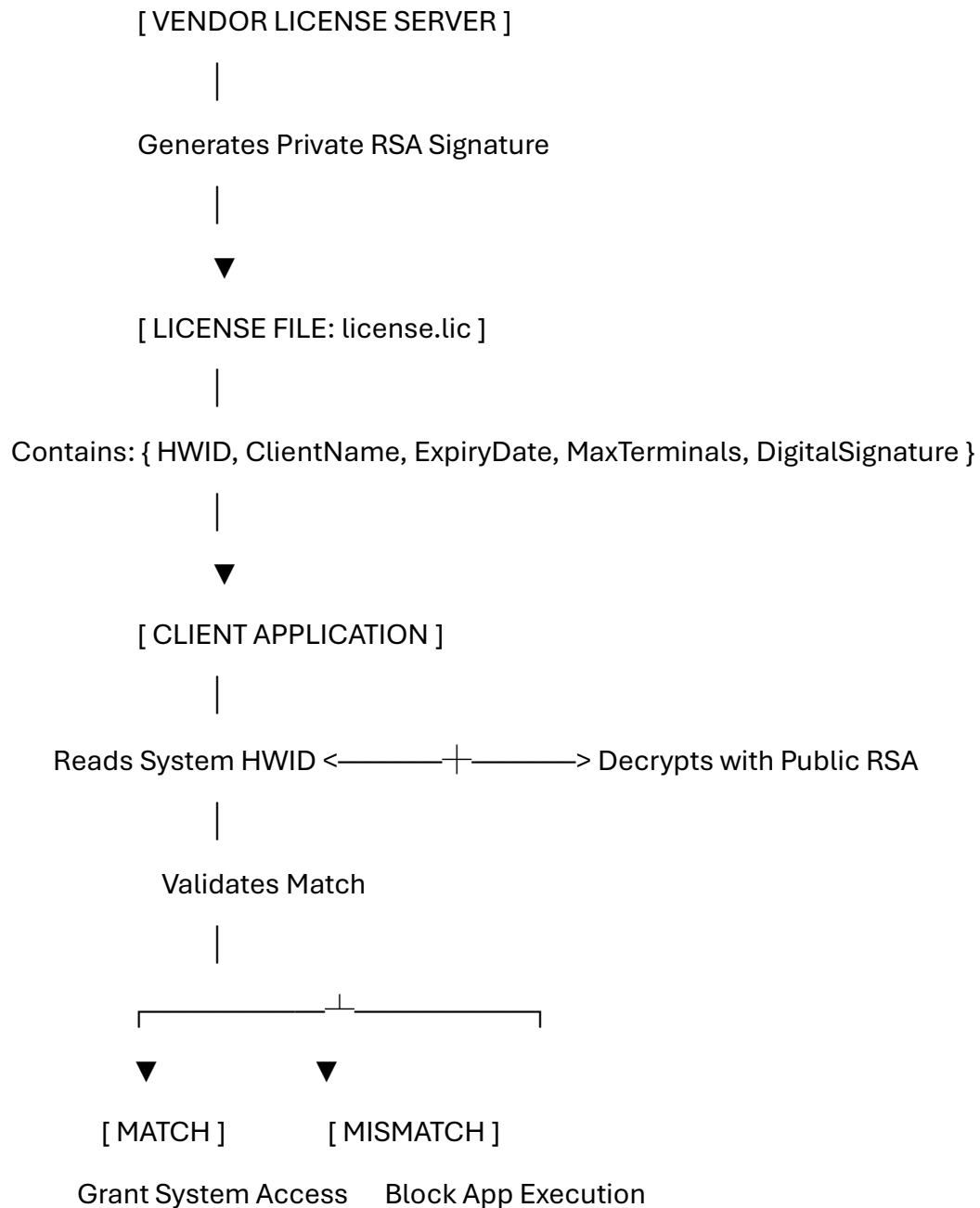
```

Barcode Label Printing

- **Formats Supported:** Code 128, QR Code generation via SkiaSharp / ZXing.
- **Label Contents:** Item Name, MRP, Expiry Date, Custom Barcode Graphic.

Part 9 — License System

Chapter 10 — Hardware License Engine Architecture



License Generation & Cryptographic Verification

1. **HWID Binding:** SHA-256 hash constructed from local hardware attributes.
2. **License File Payload (license.lic):**

JSON


```
{  
  "Licensee": "City Pharmacy Ltd",  
  "HWID": "e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855",  
  "IssueDate": "2026-08-01",  
  "ExpiryDate": "2027-08-01",  
  "GracePeriodDays": 7,  
  "Tier": "Lifetime_StandAlone",  
  "Signature": "MEQCID...=="  
}
```

3. **Computer Replacement Policy:** Re-keying requires generating a signed License Revocation Certificate on the prior machine or submitting vendor validation proof.
-

Part 10 — Backup & Disaster Recovery

Chapter 11 — Backup Architecture

- **Automated Cold Backup:** Executed every time the application is closed. The active database connection pool is drained, and an encrypted snapshot `pharma_backup_YYYYMMDD.bak` is written to a designated directory.
 - **Hot Backup:** SQLite Online Backup API is utilized if a backup is requested while the store is operating.
 - **Restoration Procedure:**
 1. Authenticate with Master Admin Credentials.
 2. Select target `.bak` file.
 3. Validate cryptographic checksum and HWID match.
 4. Replace active `pharma_db.sqlite` file and automatically reboot the system.
-

Part 11 — Non-Functional Requirements

Chapter 12 — System Quality Attributes

- **Performance:** POS Barcode Scan-to-Cart rendering response time ≤ 100 ms. Database queries respond under 50 ms for datasets up to 500,000 records.
- **Reliability:** Operational Uptime target of 99.99% for offline execution (Zero dependency on external networks).
- **Maintainability:** Modular architecture separating UI, Business Logic, Data Access (MVVM Pattern).
- **Availability:** Available 24/7/365 on local terminal hardware.
- **Usability:** 100% keyboard navigable POS system (F1–F12 shortcuts) for high-speed cashier operations.
- **Portability:** Executable on any Windows 10/11 x64 architecture with .NET 8 runtime.

Part 12 — Testing Strategy

Chapter 13 — Verification Matrix

Test Suite	Scope	Target Component	Success Criteria
Unit Testing	Individual classes	FEFO Query Logic, HWID Extractor, Tax Calculators	100% Code Path Pass in xUnit
Integration Testing	DB + Services	Purchase Order creation updating Stock Batches	Atomic DB transaction commit
License Security Testing	Cryptography	License file payload modification / File copy to different PC	App immediately locks with LIC_TAMPERED error
Hardware Testing	Peripherals	ESC/POS Receipt Printers, Barcode Scanners	Zero character drop on rapid scanning
Performance Testing	Database Scalability	Simulated 100,000 Sales Records	Cart loading time remains < 150 ms

Part 13 — Deployment

Chapter 14 — Distribution & Installation

- **Installer:** Packaged via Inno Setup / WiX Toolset producing a single PharmaSys_Setup_v1.0.exe.
 - **Prerequisites bundled:** .NET 8 Desktop Runtime x64, VC++ Redistributable.
 - **Installation Flow:**
 1. Run PharmaSys_Setup_v1.0.exe with Windows Admin rights.
 2. Select Installation Path (C:\Program Files\PharmaSys).
 3. Installer copies binary files, creates local database structure, and registers desktop shortcut.
 4. First Boot displays HWID Extractor Screen with option to import license.lic.
-

Part 14 — Maintenance

Chapter 15 — System Maintenance Protocols

- **Database Migration & Upgrades:** Handled cleanly through Entity Framework Core Migrations (EF Migrations). Migrations execute automatically during app startup after database backup creation.
 - **Bug Fix Lifecycle:** Patch releases distributed via offline update packages (.pkg) containing signed assembly diffs verified by the License Engine.
 - **Version Control Scheme:** Semantic Versioning (MAJOR.MINOR.PATCH → e.g., Version 1.0.4).
-